Multimodal Information Fusion in Finance

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Abstract

This study quantitatively analyzes the facial expressions of Federal Reserve Chairs during FOMC press conferences (2011-2021) to systematically investigate their potential impact on financial markets. we identified and quantified key emotional features including anger and fear, correlating them with market reactions in the S&P 500 index. The empirical results demonstrate that while three extracted negative emotion indicators failed to significantly predict short-term market returns, certain emotional dimensions still exhibited modest explanatory power. These findings not only reveal limitations of non-verbal emotional signals in financial decision-making, but also open new research avenues for exploring high-dimensional emotional features and heterogeneous market responses in future studies.

Introduction

In contemporary financial markets, central bank communication affects market expectations and asset pricing, with existing research focusing mainly on textual analysis of policy statements, leaving nonverbal cues like facial expressions under-explored. This study pioneer's empirical analysis of Federal Reserve Chairs' facial expressions during FOMC press conferences, quantitatively assessing nonverbal communication's direct impact on financial markets, finding effects pronounced in high-profile meetings or forward guidance discussions. Combining facial recognition with high-frequency data, it confirms nonverbal signals as an independent information channel complementing verbal communication, offering new insights into monetary policy transmission.

Solution

To assess how nonverbal cues influence financial markets, we developed a multimodal deep learning framework with four key steps:

1. Data Source Selection

Given the high market sensitivity to **FOMC** announcements and their significant impact on volatility, we developed an OpenCV-based script to extract video frames at **2-second** intervals **from 2011-2021** Fed press conferences. This enables precise facial emotion capture during critical policy communication moments.

2. Facial Expression Recognition

Using **Alibaba Cloud's FaceBody API**, we analyzed each video frame for nine emotions (anger, disgust, fear, frown, happiness, neutral, pout, sadness, surprise) with confidence scores. Image resolution was enhanced via **PIL** preprocessing, particularly for low-quality segments, to boost recognition accuracy. An automated script processed batch inputs (CSV) and exported structured results (Excel).

Below are some examples of identification:



Panel A: Ben Bernanke June 19th 2013
Emotion: Happiness
Intensity Score: 0.712890625



Panel C: Jerome Powell September 25th 2018
Emotion: Neutral
Intensity Score: 0.880371094



Panel B: Janet Yellen December 13th 2017
Emotion: Anger
Intensity Score: 0.876953125

3. Data Fusion and Feature Engineering

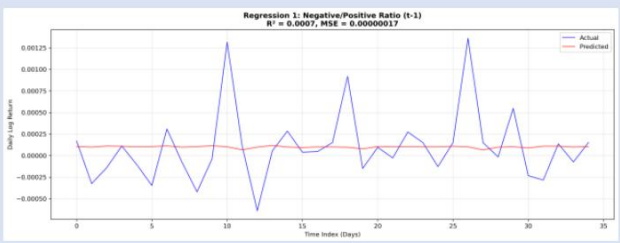
The identified emotions were mapped to emotion values and time-aligned with **10-minute return data from the S&P 500 index**. We used regression and emotions at time (t-1) as input features to predict market reactions at time t.

4. LSTM Prediction Model Construction

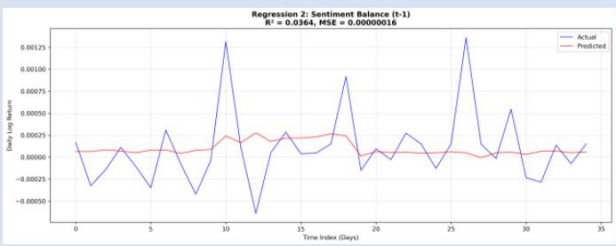
Based on the fused data, we trained **a Long Short-Term Memory (LSTM) network** to model dynamic relationships in time series. The model was evaluated using mean squared error on the test set and demonstrated good fit and predictive capability during the stable phase of 800–1000 minutes following the policy announcement.

Result

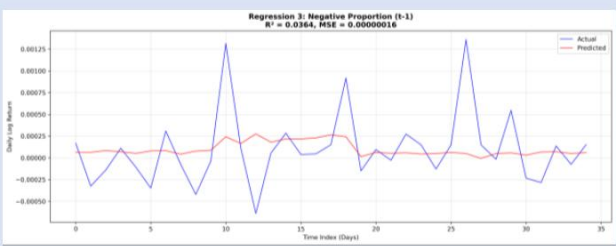
Based on facial expression recognition data, we calculated three negative emotion indicators:



Negative/Positive Ratio (t-1): The ratio of negative emotions (sadness, fear, disgust, anger) to positive emotions (happiness, surprise) in the previous period



Sentiment Balance (t-1): The standardized difference between negative and positive emotions, calculated as (negative - positive) / (negative + positive)



Negative Proportion (t-1): The proportion of negative emotions out of all emotions

The regression results reveal **limited predictive power across all emotion indicators**:

- The negative/positive emotion ratio (t-1) shows minimal explanatory power (R^2=0.0007, MSE=0.00000017).
- The Emotional Balance Index and Negative Emotion Proportion (t-1) demonstrate identical performance (R^2=0.0364, MSE=0.00000016), offering only marginally better fit.
- Visual analysis confirms these findings, showing predicted values largely tracking actual returns without capturing meaningful market movements.

Key observations:

- All models exhibit substantial overlap between predicted and actual values.
- While some indicators show **slightly better fit**, none achieve significant predictive capability.
- The consistent pattern suggests **emotional indicators have limited utility for short-term return prediction**.

Conclusion

This study aims to explore the impact of the facial expressions of the Federal Reserve Chair during FMOC from 2011 to 2021 on financial markets. The results indicate that three quantitative negative emotion indicators derived from the facial expressions of the Federal Reserve Chair during meetings do not have significant predictive power for the short-term returns of the S&P 500 Index. Although the current predictive utility is limited, sentiment analysis still holds exploratory value in financial markets. Future research directions include examining longer time horizons (such as daily or weekly returns) and developing more comprehensive sentiment indicator systems by integrating voice, text, and facial expression data. Such methodological improvements may help to more accurately identify the relationship between sentiment signals and asset price movements.

Reference

Curti, F. and Kazinnik, S. (2023) 'Quantifying the impact of nonverbal communication in FOMC press conferences', *Journal of Monetary* Economics, 139,110-126. Available at: Let's face it: Quantifying the impact of nonverbal communication in FOMC press conferences – ScienceDirect.